MoE Kernels and Routed Computation

Deep Dive into Kimi K3 — Chapter 36

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Foundation Books Project

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every selected contribution must be named, moved, and restored —
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Goal: type the routed contract, build count/scan/permute/restore and grouped SiTU experts, separate expert-centric and token-centric schedules, differentiate and replay dispatch, and keep every cost and measurement in scope.

Roadmap

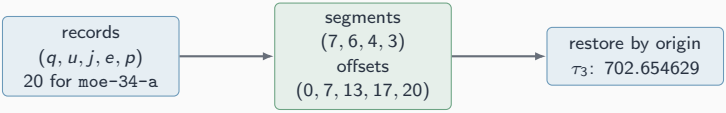
Route records and the expert-major layout

Grouped experts, fusion, and phase schedules

Backward, scheduling, and evidence

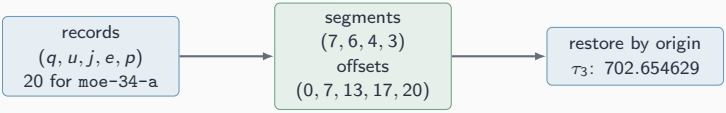
Route records and the expert-major layout

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- Permutation transparency: bijection, co-travel, expert purity, and one-time combine give the token-major output in exact arithmetic; padding stays invisible.

Grouped experts, fusion, and phase schedules

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- Legal fusion keeps every typed output: router and latent projection may share x_t; RMSNorm must follow the weighted sum.
- Prefill and training favor expert-centric grouping; small decode may favor token-centric accumulation. [REPORTED: K3 Sec. 5.4.2]

Backward, scheduling, and evidence

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Structure before values. Next: Chapter 37's MoonEP and Perfectly Balanced Expert Parallelism.